

# Production Material Dashboard

Technical and functional documentation — generated 20 August 2026

## 1. Purpose

The dashboard provides a current production overview for four product categories. Each product card shows estimated raw-material days remaining, semi-ready stock, ready stock, and current-month wastage.

## 2. Source Files

| File                             | Responsibility                                                                                                       |
|----------------------------------|----------------------------------------------------------------------------------------------------------------------|
| production_material_dashboard.py | Loads stock, consumption, and wastage data and exposes the whitelisted dashboard API.                                |
| production_material_dashboard.js | Creates the Frappe page, calls the API, renders product cards, styles status indicators, and handles refresh/errors. |

## 3. Product Configuration

| Label     | Manufacturing Type | Semi-ready Warehouse | Ready-stock Warehouse(s)                                                    |
|-----------|--------------------|----------------------|-----------------------------------------------------------------------------|
| PONDLINE  | PONDLINE           | PONDLINE FG - OM     | GD REWINDING Ready To Dispatch - OM<br>JOINT MACHINE Ready To Dispatch - OM |
| PIPE      | PIPE               | PIPE FG - OM         | SMALL PIPE M/C Ready to dispatch - OM                                       |
| PP EXPORT | PP EXPORT          | PP EXPORT FG - OM    | NAVRANG-REWINDING M/C Ready To Dispatch - OM                                |
| MURGHAS   | JP M/C MURGHAS     | MURGHAS FG - OM      | Not configured                                                              |

## 4. API and Response

The browser calls the whitelisted method:

```
hariom_sanpra.hariom_sanpra.page.production_material_dashboard.production_material_dashboard.get_dashboard_data
```

The method returns one object per configured product:

```
{
  "label": "PONDLINE",
  "icon": "drop",
  "raw_material_days": 8.4,
  "semi_ready": 12.325,
  "ready_stock": 6.150,
  "wastage_percent": 2.45
}
```

## 5. Calculations

### 5.1 Raw-material days remaining

- Find the boundaries of the `Raw Material` item-group tree.
- Read submitted Stock Entries for the default company covering the last 30 calendar days, including today.
- Include source-warehouse rows that are not finished items and whose items belong to Raw Material or a child group.
- Group consumption by manufacturing type and item.
- Retrieve current stock through the custom Stock Balance Hariom report.

Days remaining = Available raw-material quantity / (Quantity consumed during 30 days / 30)

- The backend result is rounded to one decimal place.
- Negative stock is treated as zero.

- Items without recorded consumption are excluded.
- No applicable data produces a dash in the UI.

### 5.2 Semi-ready and ready stock

Balances from the configured warehouses are obtained through Stock Balance Hariom.

```
Displayed tons = Sum of warehouse balance quantities / 1000
```

The result is rounded to three decimal places. Warehouse matching ignores case and normalizes repeated spaces or tabs. MURGHAS has no ready-stock warehouse, so the dashboard displays a dash.

### 5.3 Monthly wastage

The backend runs the Daily Cumulative Production Report from the first day of the current month through today, with the relevant manufacturing type and `Item Wise` basis. It reads `wastage_percent` from the final total row. Missing data produces `0%`.

## 6. Frontend Behavior

1. Create the Frappe page titled **Production Material Dashboard**.
2. Show a loading message.
3. Call the backend method and render one card for each result.
4. Allow manual reload using the primary **Refresh** action.

| Displayed Days | Status  | Color                |
|----------------|---------|----------------------|
| 0–3            | Danger  | Red                  |
| 4–10           | Warning | Orange               |
| More than 10   | Safe    | Green                |
| No value       | Unknown | Neutral, with a dash |

```
For display and status selection, fractional days are rounded down to a non-negative whole number.
```

The layout uses two cards per row on desktop, one card per row below 900 px, and vertically stacked metrics below 520 px.

## 7. Error Handling

If the API request fails, the loading content is cleared and Frappe shows an **Unable to load dashboard** message. Partial or cached cards are not retained.

## 8. Extension

Add a product by appending an entry to the Python `PRODUCTS` tuple:

```
{
    "label": "PRODUCT NAME",
    "icon": "box",
    "manufacturing_type": "MANUFACTURING TYPE",
    "semi_ready": ("WAREHOUSE - OM",),
    "ready_stock": ("READY WAREHOUSE - OM",),
}
```

Supported icons are `drop`, `pipe`, `box`, and `layers`. Unknown values use the box icon.

## 9. Operational Assumptions

- A default company is configured.
- `custom_manufacture_type` is populated on relevant Stock Entries.
- Raw materials belong to Raw Material or one of its descendant groups.
- Dividing stock quantity by 1,000 correctly converts the relevant stock unit into tons.
- Raw-material quantities added together use compatible units of measure.
- Users have permission to open the page and access the underlying reports and records.

